Internship Overview

Autonomous Weeds and Terrain Perception

[ID 012026]

Computer vision and ranging pipeline for field robotics support.

Objectives

- Improve autonomous navigation and vegetation monitoring for mobile platforms.

Tasks

- Detect invasive weeds and tag them
- Approximate the grass height
- Detect obstacles and calculate approximate distance

Requirements

- Python programming
- Computer vision fundamentals
- Linux command line experience

Equipment

- Jetson Orin Nano
- Arducam IMX129
- VL53L0X ranging sensors
- LED 4" Auxbeam
- Teltonika RUT240 Router

Deliverables

- Prototype perception software stack with labeled detections and distance estimates

Work Breakdown

- 20% hrdware integration. 80% coding

Duration: 2 months

Payment: Available in Bizquick

Supervisor: Kaleb Granados